

Practical No. 04

ON

SSCA4030 - Image Processing

TITLE: Apply Spatial Filters (Mean, Median, Gaussian, and Laplacian) Using Python and OpenCV

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

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Aim: To apply different spatial filters (Mean, Median, Gaussian, and Laplacian) on a grayscale image using OpenCV and compare their results.

Input Code:

```
import cv2
from matplotlib import pyplot as plt

# Read grayscale image
img = cv2.imread("/content/RISHABH_15.png", cv2.IMREAD_GRAYSCALE)

# Apply a 3×3 Mean (Average) Filter on the input image
# Mean Filter
mean = cv2.blur(img, (3, 3))

# Median Filter
median = cv2.medianBlur(img, 3)

# Apply a 3×3 Gaussian Filter
# Gaussian Filter
gaussian = cv2.GaussianBlur(img, (3, 3), 0)

# Laplacian Filter (with larger kernel size)
laplacian = cv2.Laplacian(img, cv2.CV_64F, ksize=5)
laplacian_abs = cv2.convertScaleAbs(laplacian)

# Apply threshold to make edges more visible
# The threshold value (50) may be adjusted depending on the image
_, laplacian_thresholded = cv2.threshold(
    laplacian_abs,
    50,
    255,
    cv2.THRESH_BINARY
)
```

```
# Display Results
```

```
titles = [
```

```
    "Original",
```

```
    "Mean Filter",
```

```
    "Median Filter",
```

```
    "Gaussian Filter",
```

```
    "Laplacian Filter",
```

```
    "Laplacian Filter "
```

```
]
```

```
images = [
```

```
    img,
```

```
    mean,
```

```
    median,
```

```
    gaussian,
```

```
    laplacian_abs,
```

```
    laplacian_thresholded
```

```
]
```

```
plt.figure(figsize=(15, 10))
```

```
for i in range(len(images)):
```

```
    plt.subplot(2, 3, i + 1)
```

```
    plt.imshow(images[i], cmap="gray")
```

```
    plt.title(titles[i])
```

```
    plt.axis("off")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:

Original



Mean Filter



Median Filter



Gaussian Filter



Laplacian Filter



Laplacian Filter

